Jothi Baskar

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EDUCATION

Vellore Institute of Technology (VIT) **2027**
B Tech in Computer Science Engineering with Specialisation in Internet of Things (IoT) *Vellore, India*
CGPA: 8.43

Narayana CO School **2023**
Higher Secondary School *Bangalore, India*
89.20%

Delhi Public School Bangalore East **2021**
Secondary School *Bangalore, India*
93.17%

PROJECTS

IoT-Based Smart Parking System
NodeMCU ESP8266, MQTT, Embedded C/C++, IoT, Sensor Networks

- Developed a smart parking solution using NodeMCU ESP8266, IR sensors, and MQTT protocol for real-time parking occupancy monitoring and remote status updates.
- Implemented automated gate control using sensor-driven decision logic and servo actuation based on vehicle detection and slot availability.
- Designed a layered IoT architecture integrating sensor interfacing, wireless communication, embedded systems, and real-time monitoring through LCD and mobile dashboards.

Fog-Assisted Real-Time Shuttle Tracking System for Smart Campuses
Python, IoT, Edge Computing, Fog Computing, REST APIs, JavaScript

- Built a distributed Edge–Fog–Cloud architecture for real-time shuttle tracking, enabling low-latency telemetry processing and scalable campus mobility management.
- Developed a topological path-summing ETA algorithm using route-aware distance calculations and Haversine-based geospatial computations for accurate arrival predictions.
- Implemented GPS interpolation, RESTful communication, and SVG map-snapping techniques to deliver smooth shuttle visualization while reducing latency by 72% and network traffic by 60%.

VOLUNTEERING

Samprapti Foundation **December 2025**

- Completed 6 weeks of voluntary service at Samprapti Foundation, supporting elderly residents through meal service, daily assistance, and meaningful social interaction while contributing to a caring and inclusive environment.

SKILLS & INTERESTS

- Technical Skills: Java, C, Python, Data Structures & Algorithms, Internet of Things (IoT), Embedded Systems Fundamentals, Sensor Interfacing.
- Design Skills: Wireframing, Website Layout Design, Color Theory, Visual Design Principles, User-Centered Design
- Professional Skills: Analytical Problem-Solving, Adaptive Learning, Team Collaboration, Presentation Skills.
- Interests: UI/UX Design, Digital Art, Sketching, Mural Painting, Robotics & Emerging Technologies.